The Production Linux Troubleshooting Cheat Sheet

Run these commands inside your remote Azure VM terminal session (`azureuser@Matecca-VM:~$`) or check these local configurations to inspect, revive, or debug any slice of your CRM stack.

🐍 1. Python Flask API Layer (systemd)

- `sudo systemctl status flask-crm.service`
  - ***What it does:*** *Checks if your Python API is alive, dead, or throwing errors. Look for a green `active (running)` tag.*
- `sudo systemctl restart flask-crm.service`
  - ***What it does:*** *Restarts your Python backend. Run this anytime you modify your code inside `main.py`.*
- `tail -f /home/azureuser/python_api/flask.log`
  - ***What it does:*** *Displays your API logs live. Leave this open in your terminal while submitting forms to watch inbound payloads and data errors scroll by in real-time. (Press **Ctrl + C** to exit).*
- `sudo killall -9 python` *or* `sudo killall -9 python3`
  - ***What it does:*** *Terminates every running Python process on your system at once if a ghost loop freezes port 8000.*
- `sudo lsof -i :8000`
  - ***What it does:*** *Verifies if port 8000 is empty or tells you exactly what process ID (PID) is currently locking it up.*

🌐 2. Nginx Transport Layer

- `sudo nginx -t`
  - ***What it does:*** *Sanity check for configuration syntax. Run this before restarting Nginx to catch missing semicolons or typos.*
- `sudo systemctl status nginx`
  - ***What it does:*** *Checks if Nginx is active on Port 443.*
- `sudo systemctl restart nginx`
  - ***What it does:*** *Gracefully restarts your proxy engine to load changes made within `sites-available/matecca-crm-proxy`.*
- `sudo tail -f /var/log/nginx/error.log`
  - ***What it does:*** *Prints your public server's network routing error streams if your frontend form completely fails to reach the VM.*

⚠️ ***CRITICAL ENTERPRISE BUG:*** *By default, Nginx strips out all custom HTTP headers that contain underscores (like* `X-Matecca-API-Key`*), throwing automatic* ***401 Unauthorized*** *errors.*

The Fix: *Ensure* `underscores_in_headers on;` *is explicitly added inside the main* `server {}` *block of your* `/etc/nginx/sites-available/matecca-crm-proxy` *profile script!*

🐘 *3. Apache Dashboard Layer*

- `sudo systemctl status apache2`
  - ***What it does:*** *Confirms if your Apache presentation layout is listening safely on Port 8080.*
- `sudo systemctl restart apache2`
  - ***What it does:*** *Restarts Apache. Use this after changing virtual directories or updating dashboard PHP layouts.*

🗄️ *4. MySQL Relational Database Layer*

- `sudo systemctl status mysql`
  - ***What it does:*** *Confirms that the database daemon is active.*
- `mysql -u crm_user -p`
  - ***What it does:*** *Logs you into the MySQL shell. It will securely prompt for* `Matecca_Secure_2026!`*.*
- `USE saas_crm; SELECT * FROM crm_leads;`
  - ***What it does:*** *(Run inside the MySQL shell) Displays every single enriched record saved to your cloud schema so you can verify that data writes are functional.*

🛠️ ***LOCAL CONFLICT ERROR:*** *“Unable to start site connect to server failed”*

The Fix: *Open Windows Task Manager on your local PC, search for any lingering* `mysqld.exe` *processes under background tasks, and force end them to stop local port conflicts with your cloud testing stream.*

📡 *5. Azure Network & VM Connectivity*

- ***IP Address Synchronization Validation:***
  If your machine is stopped and restarted in the Azure Portal, its public IP can change. ***Always verify*** *that your current active portal IP exactly matches the target string set inside your local WordPress core environment (`$secure_azure_url`).*
- ***SSH Remoting Login Target Line:***
  Open your PC's Command Prompt (CMD) and execute:
  `ssh -i "C:\Users\ramen\OneDrive\Desktop\Project\Matecca-VM_key.pem" azureuser@172.208.96.146`
  (Change path mapping entries manually if the private `.pem` credential key asset is moved).

🧩 *6. WordPress Frontend Gateway Layer*

- ***The "Security Error: Gateway Connection Fault" Cache Loop:***
  If you edit script text strings or styles but the frontend browser layout layout displays old behavior, it means code properties are cached or stuck in a database block.
  1. ***Clear Ghost Scripts:*** *Check page templates or structural block setups for old `<script>` blocks and delete them. Never embed scripts directly into the WordPress post editor.*
  2. ***Break Browser Cache:*** *Press* ***Ctrl + F5*** *inside Firefox to clear physical file memory blocks.*
  3. ***Step Up Asset Versions:*** *Change version properties directly inside `wp_register_script()` or `wp_enqueue_style()` (e.g., bump `'2.2.0'` to `'2.2.1'`) to force the local server to break old cache lines.*
- ***cURL Error 28 (Connection Timeout) Network Block:***
  If submissions throw a `Timeout was reached` warning, the connection is dropping at the cloud perimeter firewall.
  1. *Log into the* ***Azure Portal*** *➔ Go to* ***Network settings*** *for your VM.*
  2. *Click* ***Create port rule*** *➔* ***Inbound port rule****.*
  3. *Select* ***HTTPS*** *as the service, set the destination port to* ***443****, and configure a unique priority value like* ***305****. Click* ***Create****.*
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